Journal of Asian Economics 98 (2025) 101926

Contents lists available at ScienceDirect

Journal of Asian Economics

journal homepage: www.elsevier.com/locate/asieco

The impact of artificial intelligence on labor market: A study based on bibliometric analysis

Huilin Zhou [a], Linhui Wang [a,*], Yutong Cao [b], Jincheng Li [c]

[a] School of Economics and Management, East China Normal University, China
[b] School of Business and Management, Jilin University, China
[c] Center for China Public Sector Economy Research, Jilin University, China

ARTICLE INFO

JEL classification:
J21
J23
O14
O33

Keywords:
Artificial intelligence
Labor market
Bibliometric analysis
Systematic overview

ABSTRACT

The impact of artificial intelligence (AI) on the labor market has become a critical focus of academic research. However, most studies only focus on single dimensions or perspectives, with few efforts to systematically synthesize key insights from frontier research on AI's effects on the labor market. To address this, we use the CiteSpace to analyze 1647 articles in the Web of Science between 2007 and 2024 to identify research hotspots and the evolution of themes. Then, we further illustrate the representative views of AI on the labor market from five perspectives: employment, skills, job quality, wage and policy. The findings are as follows: (1) Since 2015, research on AI and the labor market has expanded rapidly, particularly in areas such as labor participation, wage inequality, job stability and the development of AI technologies. (2) Bibliometric analysis reveals significant focus on the effects of AI on employment and skill demand, with "digital workplace" "job insecurity" "job quality" and AI applications emerging as major research hotspots. (3) We provide a systematic overview on AI and employment, skills, job quality, wages and policies, and present representative views. (4) Future research may focus on the impact of AI on job quality, emerging forms of work and the varying impact of different AI technologies on the labor market.

1. Introduction

The widespread application of AI is driving a transformative shift toward intelligent and automated production processes, significantly improving efficiency across various industries. Within the domain of manufacturing, intelligent devices enhance productivity and reduce the occurrence of human errors (Li et al., 2020). In the consumer market, AI enables precise demand forecasting through data-driven predictions and dynamic pricing (Abrardi et al., 2022). In the financial sector, AI optimizes credit approval, investment decisions, and fraud detection through intelligent risk assessment and data analytics, enhancing the security and accuracy of financial services (Durongkadej et al., 2024; Pan et al., 2024).

The transformative impact of AI is reshaping the labor market and has become a focal point of academic research. Current studies extensively examine its impact on employment (Acemoglu & Autor, 2011; Acemoglu & Restrepo, 2018; Bessen et al., 2023), skill demand (Goos et al., 2014; Xie et al., 2021), wage distribution (Acemoglu & Restrepo, 2022), and shifts in employment structures

* Corresponding author.
E-mail addresses: zhouhl555@163.com (H. Zhou), linhuiwang73@sina.com (L. Wang), ytcao24@mails.jlu.edu.cn (Y. Cao), jinchengli86@163.com (J. Li).

https://doi.org/10.1016/j.asieco.2025.101926
Received 19 January 2025; Received in revised form 12 March 2025; Accepted 27 March 2025
Available online 29 March 2025
1049-0078/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Table 1
Top journals in AI and labor market research.

Journals	Count
Technological Forecasting and Social Change	57
Technology in Society	41
Journal of Business Research	26
Research Policy	17
Labour Economics	17
Humanities & Social Sciences Communications	16
Applied Economics	16
Structural Change and Economic Dynamics	15
Economics Letters	14
Journal of Economic Behavior & Organization	12

3. Bibliometric analysis and results

Bibliometric analysis visualizes research development, revealing hotspots and evolutionary trends. In this section, we use the CiteSpace to analyze literature from the Web of Science database, deepening the understanding of key research areas and their evolution in AI and the labor market.

3.1. Research hotspots

Keywords reflect the core themes of research literature. Using CiteSpace for co-occurrence and clustering analysis, along with knowledge mapping, helps identify key research hotspots and highlights differences in focus across themes, providing a systematic understanding of the research focus on AI and the labor market.

3.1.1. Keyword co-occurrence analysis

Fig. 2 presents the keyword co-occurrence knowledge map by CiteSpace. Specifically, data from the Web of Science database were processed to construct a co-occurrence network based on keywords. In the map, each node represents a keyword, with node size indicating its frequency of appearance. The centrality of a node reflects its importance in linking related or extended keywords within the network. As depicted in Fig. 2, the keyword co-occurrence network reveals tightly connected relationships and clear co-occurrence patterns. "Artificial intelligence" emerges as the most frequently occurring keyword, underscoring its significant impact on the labor market. High-density terms linked to it, such as "technology" "technological change" "employment" "jobs" "growth" "performance" and "future" highlight key research hotspots. Based on these keywords, we further categorized the research into three parts for further analysis.

The first part focuses on the relationship between AI and employment, as shown in the lower section of Fig. 2. Keywords such as "artificial intelligence" "employment" "jobs" and "labor" have the largest nodes and highest centrality, reflecting a concentrated focus on AI's impact on employment. Over time, studies on the employment effects of AI have become more nuanced, with terms like "job polarization" highlighting the unequal distribution of job opportunities, and keywords such as "skill" "wage inequality" and "unemployment" indicating interest in skill shifts, workplace inequality and unemployment. Additionally, emerging topics include job

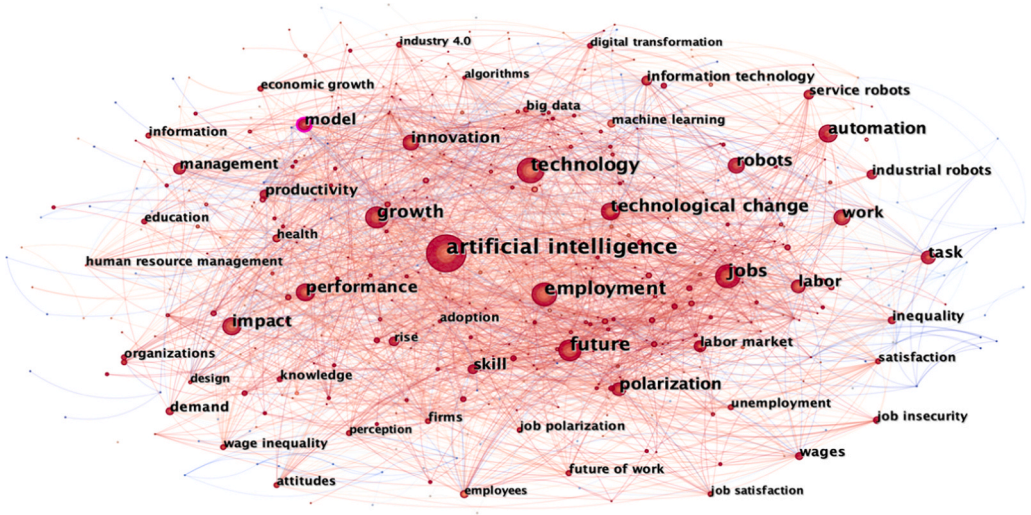

Fig. 2. Keywords co-occurrence.

stability under technological change, as evidenced by keywords like "job satisfaction" "job insecurity" and "future of work". Although these nodes remain smaller, they suggest early-stage exploration and potential avenues for future research.

The second part centers on technological change and AI development, located in the upper-right section of Fig. 2. High-frequency keywords such as "technology" and "technological change" emphasize the critical role of technological progress in shaping the labor market. Additionally, varied technological changes at different stages exert distinct impact on labor market. Between 2007 and 2024, keywords like "information technology" "automation" and "robots" highlight key aspects of technological progress. "Information technology" has facilitated "digital transformation" through internet platforms, while robots and automation have reshaped the traditional labor market supply-demand structure. Notably, "robots" as major embodiment of AI technologies has drawn attention in subfields such as "industrial robots" and "service robots." Additionally, with the rise of "big data", research on "algorithms" and "machine learning" has become more refined and gained significant traction.

The third part highlights specific applications and industry domains of AI, located in the upper-left section of Fig. 2. Keywords such as "innovation" "productivity" and "growth" reflect AI's positive impact on productivity and economic growth. High-frequency terms like "management" "Industry 4.0" and "education" illustrate AI's diverse applications. In human resource management, AI-enabled remote interview platforms reduce recruitment costs and improve efficiency. In smart manufacturing, intelligent devices and automated machinery enhance production and management processes. Similarly, in education, AI-powered assessment systems autonomously evaluate student performance and learning progress.

3.1.2. Keyword cluster analysis

To analyze the distribution of themes within literature on AI and the labor market, Fig. 3 presents a clustering map by the online text analysis tool Carrot2. Carrot2 categorizes the abstracts of 1647 articles from the Web of Science database into clusters based on textual similarity. Each cluster is labeled with the three most relevant terms and includes at least 20 articles, forming 24 clusters shown on the left of the figure, with line length indicating the number of articles in each category. These clusters are further grouped into four technology categories, with colors ranging from dark to light representing AI-related, robots-related, automation-related and others. Most clusters are AI-related and robots-related, reflecting the widespread use of robots as a prominent AI application in production. Moreover, the clusters "Automation, Labor, Productivity" "AI, Automation, Skills" and "Work, Digital, Technology" contain the largest number of articles, focusing on automation, digital technologies, and their effects on labor productivity, employment, and skills. While the figure reflects the thematic diversity within the dataset, the labels based on only three keywords are limited in their explanatory power, necessitating further analysis to fully capture the underlying research themes.

To more accurately identify research themes on AI and the labor market, we visualized the research themes by using the keyword clustering function in CiteSpace, resulting in eight clusters shown in Fig. 4. Further, we grouped these clusters into four categories for detailed analysis.

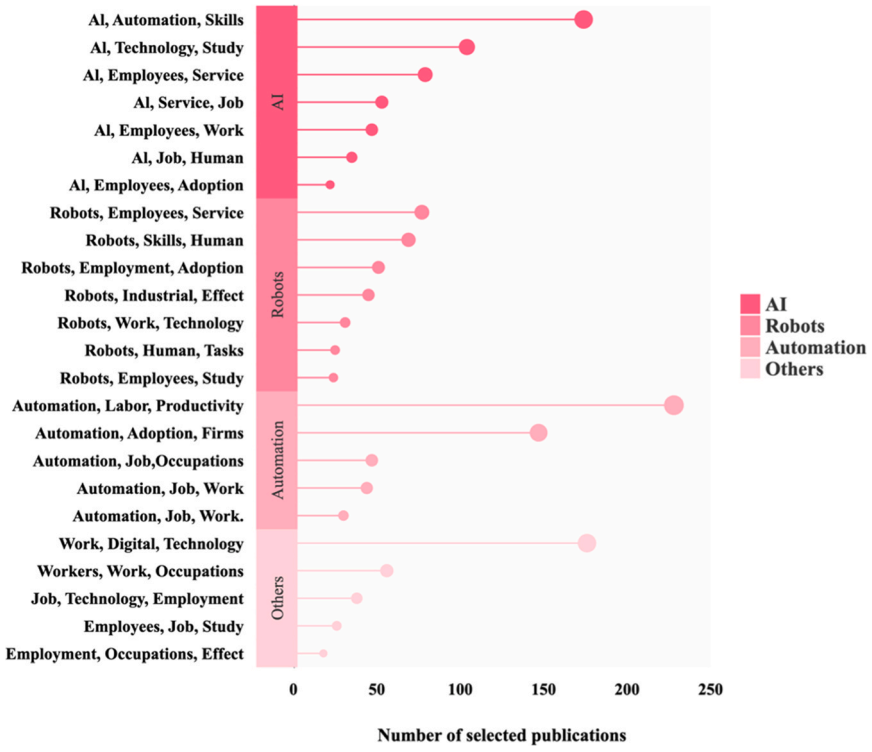

Fig. 3. Keywords cluster: based on Carrot2.

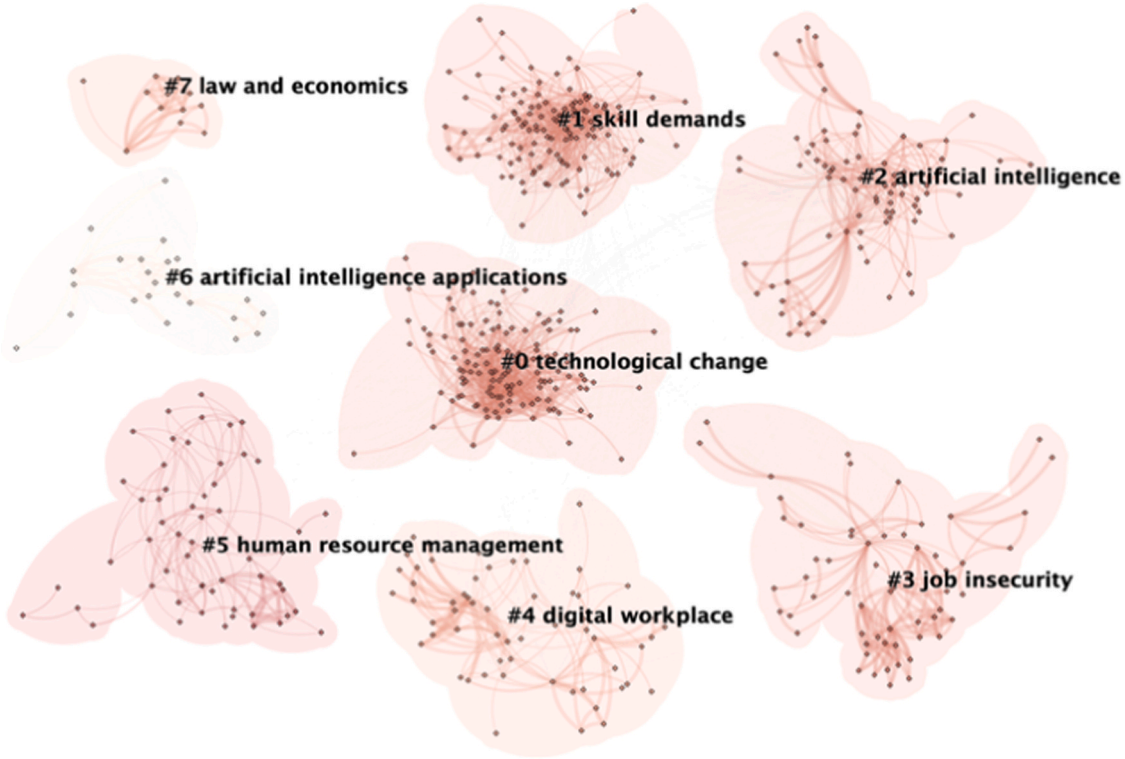

Fig. 4. Keywords cluster: based on CiteSpace.

The first group comprises the clusters "technological change" and "artificial intelligence," focusing on the evolution of technological progress in the labor market. As an emerging technology, AI plays a pivotal role in transforming production processes, driving industrial change, and reshaping employment patterns, making it a central theme in studies of technological progress.

The second group includes the clusters "skill demands," "digital workplace," and "job insecurity," primarily addressing the impact of AI on occupations and individuals. At the occupational level, "skill demands" highlights that AI changes occupational tasks and

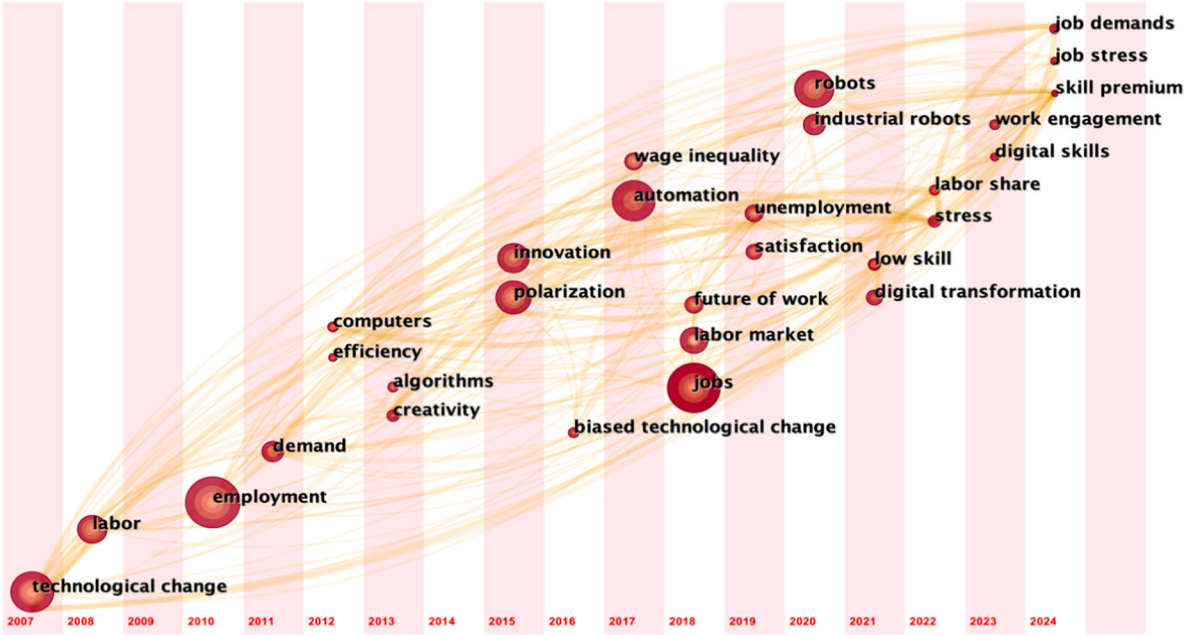

Fig. 5. Keywords timezone.

requires various skills, a widely studied theme. Meanwhile, research on the “digital workplace” remains limited, suggesting potential for future exploration. At the individual level, topics such as unemployment risks from intelligent technologies and strategies to cope with technological shocks have emerged as critical research issues.

The third group includes the clusters “human resource management” and “artificial intelligence applications”, focusing on the practical applications of AI in the labor market. These clusters cover areas such as recruitment, performance evaluation and employee training, emphasizing the growing role of AI in human resource management.

The fourth group, represented by the cluster “law and economics”, focuses on the legal and economic implications of AI. As AI-driven transformations reshape labor markets, the development of appropriate legal frameworks and economic policies has become increasingly critical.

3.2. Trends in the evolution of research hotspots

To analyze the dynamic changes in research hotspots, we used the CiteSpace to generate the keywords timezone graph in Fig. 5 and keywords burst detection in Table 2. The timezone graph shows the keywords distribution over time, with the horizontal axis representing year and the yellow curve indicating co-occurrence of two keywords in the same articles. The keywords burst strength highlights the impact of keywords during specific periods, with higher strength indicating greater influence, while duration represents the persistence of each hotspot.

To ensure the clarity of the analysis, irrelevant or low-impact keywords were excluded from the study. The sample was divided into two phases, 2007–2014 and 2015–2024, in conjunction with the number of publications, to analyze the evolution of research on AI and the labor market.

Between 2007 and 2014, the number of literatures on AI and the labor market remained low, with themes primarily focused on "technological change" and "employment". Studies during this period mainly explored the impact of technological progress on the labor market, with particular attention to the potential effects of robots on employment, but lacked in-depth analysis of mechanisms, industry-specific differences, and structural changes in the labor market. In the burst detection graph, keywords such as "robots" "digital economy" "human-robot collaboration" and "algorithms" emerged in 2008, 2009, 2011, and 2013 respectively, and lasted until 2019. These bursts highlight key trends during this phase: (1) The emergence of "robots" in 2008 indicates growing concern over the job displacement effects of robots. (2) "Human-robot collaboration" in 2011 acknowledges the dual impact of robots, not only replacing jobs but also enhancing productivity through human-machine interaction. (3) The appearance of "algorithms" underscores the significant role of AI algorithms in shaping emerging employment patterns and labor market.

From 2015 to 2024, research on AI and the labor market entered a more multidimensional and in-depth phase. High-frequency keywords such as "jobs" "polarization" "labor market" "wage inequality" "automation" and "robots" indicate that research increasingly focused on the effects of AI and robots on job polarization and wage inequality. Unlike earlier macro-level analyses, this period shifted toward micro-level and explored how AI leads to job polarization, wage inequality and skill premiums. Specifically, the emergence of "wage inequality" in 2017 and "jobs" and "polarization" in 2018 reflect growing concerns over the inequality between low-skilled and high-skilled workers. By 2019, "technological unemployment" emerged as a key topic, emphasizing unemployment risks and career transition challenges faced by low-skilled workers. In 2021 and 2022, the scope of research expanded further, with increasing attention to how AI affects "job quality" and "mental health". These studies reveal that AI has changed workplaces and conditions and has also caused anxiety and affected workers’ job satisfaction and mental health due to technological changes.

Moreover, Fig. 6 also illustrates the dynamic evolution of research themes in each cluster from 2007 to 2024. Since 2007, research on "technological change" has remained consistently active. After 2015, studies on the cluster "artificial intelligence" and the labor market gradually increased. With the widespread application of AI in production, researchers began focusing on the profound impact of this emerging technology on employment, skills and wages. Then, after 2020, research on the cluster "skill demands" gained prominence, emphasizing not only job substitution and occupational disappearance but also shifts in skill requirements. Additionally, other themes such as "digital workplace" "job insecurity" "human resource management" "artificial intelligence applications" and "law

Table 2
Burst detection of keywords.

Keywords	Burst begin	Burst end	Duration	Strength
robots	2008	2015	8	1.99
digital economy	2009	2019	11	3.99
human-robot collaboration	2011	2019	9	3.30
algorithms	2013	2019	7	2.11
wage inequality	2017	2019	3	2.88
labor market	2018	2019	2	5.70
human resource management	2018	2020	3	3.97
big data	2018	2021	4	3.75
job polarization	2018	2020	3	2.99
future work	2018	2020	3	2.03
technological unemployment	2019	2021	3	4.28
labor demand	2020	2024	5	2.75
job quality	2021	2022	2	3.91
mental health	2022	2024	3	1.90

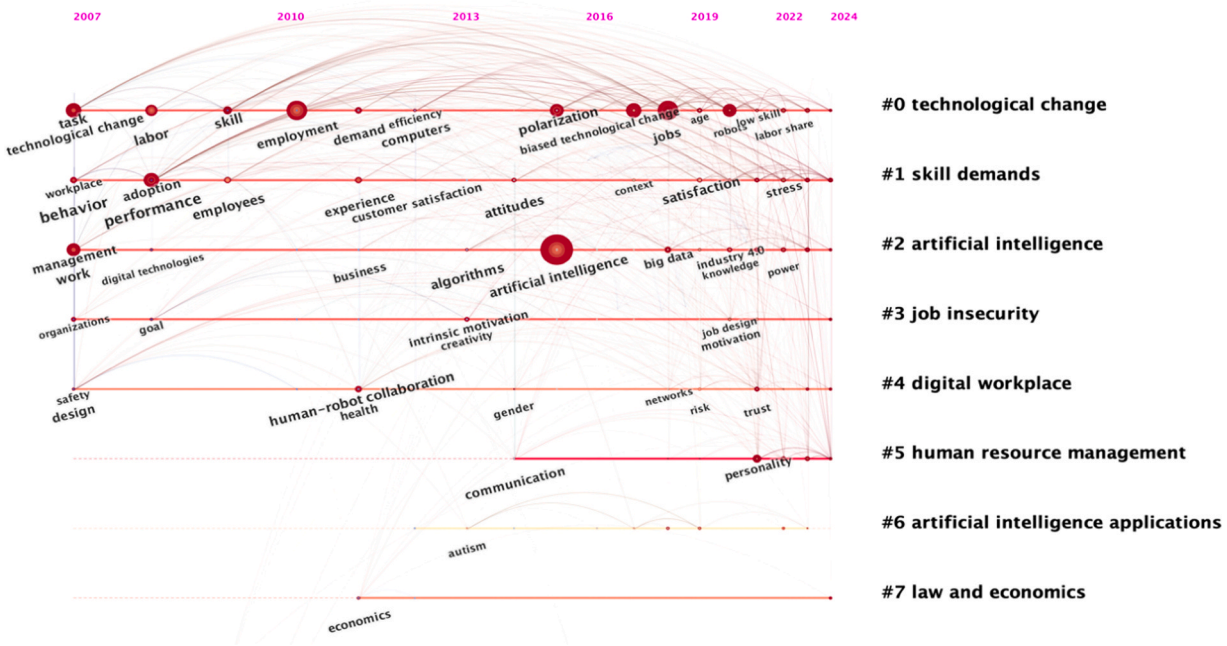

Fig. 6. Keywords timeline.

and economics" have gradually attracted attention since 2018. Although the number of publications on these topics remains small, they reflect increasing interest in how AI is changing work patterns, job stability and practical applications. The cluster "law and economics" indicates emerging concerns around social security, policy development and legal frameworks in the context of AI applications.

4. Frontier research and representative views

Based on the bibliometric analysis, we further summarized representative perspectives on AI and the labor market, focusing on five frontiers: employment, skills, job quality, wages and policy, using the 1647 articles obtained from the Web of Science, as presented in Fig. 7.

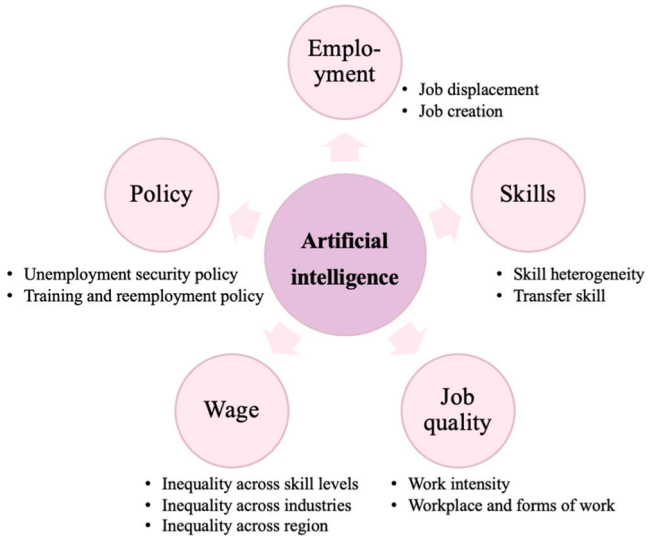

Fig. 7. Frontier research of the impact of artificial intelligence on labor market.

4.1. AI and employment

The impact of AI on employment has generated significant debate. Some argue that AI may reduce jobs by replacing workers in repetitive tasks, leading to the decline of certain occupations. Others hold that AI could increase employment by creating new tasks and driving productivity growth.

4.1.1. Job displacement

The substitution effect of AI technologies in production has drawn increasing attention (Jestl, 2024; Nguyen & Vo, 2022). Empirical evidence indicates that the adoption of robots and intelligent devices in developed regions like the United States, the United Kingdom, and Europe has reduced labor demand (Chen et al., 2022; Gravina & Pappalardo, 2022). Similarly, in developing countries such as China, Bolivia, and Chile, the substitution effect has also raised concerns about technological unemployment (Egana-delSol et al., 2022; Wang et al., 2023). These findings suggest that the displacement effects of AI spread around the world, and are more pronounced in labor-intensive industries like manufacturing. For instance, Stemmler (2023) found that industrial robots significantly reduced employment in manufacturing but increased jobs in the mining industry, exposing workers to the risk of structural unemployment. In fact, substituted jobs often involve routine, repetitive and programmable tasks, where intelligent machines outperform workers in efficiency and accuracy (Cortes et al., 2017). Gong and Png (2024) observed that automation in supermarkets reallocated tasks between cashier and machines, increasing task specialization and improving scanning efficiency by over 10 %. And jobs in fields such as accounting, sales, logistics, and trade have been increasingly replaced by specialized software and AI (Acemoglu & Restrepo, 2019). Moreover, the displacement effects are now extending to non-routine cognitive tasks, as text generation, data analysis, and visual content creation are increasingly being handled by intelligent systems (Garg et al., 2024).

4.1.2. Job creation

The rapid development of AI technologies has significantly contributed to employment growth by creating new tasks and reshaping industrial structures (Damioli et al., 2023). First, AI-driven innovation has significantly enhanced firm productivity, indirectly boosting employment through scale expansion or new business development (Qin et al., 2022). Camiña et al. (2020) found that robots and flexible production systems improve long-term productivity, reinforcing positive employment trends over time. Second, while substituting labor, AI also generates numerous new tasks. These tasks include programming, designing, and maintaining high-tech equipment, such as software and application development, database design and analysis, and computer security-related tasks, as well as more specialized roles within existing professions, such as administrative assistants, loan application analysts, and medical equipment technicians (Acemoglu & Restrepo, 2019; Lin, 2011). In addition, AI algorithms iterate rapidly, and some new occupations arise, such as intelligent vehicle testers and industrial Internet technicians. Generative AI is driving the growth of occupations related to the digital economy. It has created numerous new occupations that demand both creativity and technical expertise by optimizing content generation, improving processes, and enhancing customer experiences (Giordano et al., 2024; Xie et al., 2021). Finally, technological innovation has driven the development of emerging industries, expanding employment opportunities. In traditional industries like manufacturing, the application of intelligent devices creates employment opportunities in AI device maintenance and industrial automation management (Huang, 2024). Simultaneously, AI has restructured business models in traditional sectors, increasing demand for digital skills and creating new positions such as smart marketing specialists and technical support engineers (Duan et al., 2024; Mann & Püttmann, 2023).

4.2. AI and skills

AI has also raised concerns about shifts in skill demand. AI has altered production task allocation, reducing demand for the low-skilled while increasing demand for the high-skilled. Concurrently, the need for new skills has grown, with transferable skills becoming crucial for workers to adapt to new roles in the evolving technological landscape.

4.2.1. Skill heterogeneity

The widespread adoption of AI has significantly increased the risk of job displacement for medium -skilled and low-skilled workers. Studies have shown that AI technologies have led to a decline in medium-skilled jobs, while polarizing the labor market by increasing demand for both low-skilled and high-skilled positions (Fierro et al., 2022; Tyson & Zysman, 2022). This phenomenon is known as "job polarization" and demonstrates that medium-skilled workers are increasingly replaced by intelligent technologies. Additionally, some studies also suggest that AI-driven industrial upgrading may create opportunities for medium-skilled jobs but significantly reduce employment for low-skilled workers (Ma et al., 2022). In short, the adoption of efficient algorithms and automated equipment compresses employment choices for low-skilled workers. Lu et al. (2018) found that large-scale automation investments in healthcare services have reduced labor demand, with high-end nursing homes experiencing a 5.8 % decline in demand for registered nurses. Moreover, due to limited skills and challenges in skill transformation, low-skilled workers face declining competitiveness in the labor market, with the negative impact of AI persisting over time. For example, telephone operators displaced by automation not only declined in number but were also more likely to transition to low-paying jobs or exit the workforce entirely (Feigenbaum & Gross, 2024).

While replacing traditional jobs, AI has also increased demand for high-skilled workers. As automation develops, traditional jobs continue to decline, while the demand for knowledge-intensive jobs increases. Choi and Leigh (2024) found that industries with higher AI adoption in U.S. metropolitan cities experienced rising demand for AI-related positions but a decline in non-AI positions. In practice,

the commercialization of artificial intelligence has rapidly boosted the demand for high-skilled occupations, including senior data analysts, AI engineers, and algorithm architects. This trend is particularly significant in sectors such as healthcare, financial, and new energy sectors. For example, AI-assisted diagnostics and algorithmic trading have made high-skilled workers essential for driving innovation (Sachan et al., 2024). Moreover, AI adoption has increased demand for interdisciplinary talents, and workers with expertise in fields like computer science and economics has become popular. The demand for AI skills has risen in a wide range of occupations. Acemoglu et al. (2022) analyzed online job vacancy data and found a rapid rise in demand for AI-related skills across fields like engineering, analytics, marketing, finance, and IT.

4.2.2. Transfer skill

In the context of AI reshaping the labor market, transferable skills are essential for workers to adapt to new jobs, including communication, collaboration, and complex problem-solving. Sen et al. (2022) found that AI has altered job structures across industries, and workers with diverse skills show greater learning efficiency in human-machine interaction environments, thereby enhancing their employability. Taking the service sector as an example, employees who can efficiently collaborate with teams and customers with the support of AI system tend to perform better in their roles (Gong & Png, 2024). In general, new technological environments demand that workers possess strong communication and collaboration skills, critical thinking, creative problem-solving, content management and cross-disciplinary learning capabilities (Audrin et al., 2024). Arendt et al. (2023) analyzed over 2.3 million Polish online job vacancies and found that cognitive, communication, adaptability and self-organization skills are critical regardless of changes in job task portfolios, and communication as well as cognitive skills significantly enhance workers' employability in both routine and non-routine cognitive occupations. Additionally, AI has facilitated the emergence of new forms of work. With the rise of flexible forms of work, transferable skills help workers access to more job opportunities (Deming, 2017). Workers with project management and leadership skills often show greater adaptability during job transitions (Wang et al., 2024).

4.3. AI and job quality

The widespread adoption of AI across various production scenarios has transformed the work environment and job experiences. Research turns to focus more on job quality, including work intensity, and the change of workplace and forms of work.

4.3.1. Work intensity

With the application of AI, work patterns are becoming more flexible, and work intensity is decreasing. Companies introduce AI systems to analyze production demand and resource allocation, thereby reducing repetitive work and shortening employees' working hours (Robinson et al., 2020). Kim et al. (2023) categorized firms into labor-intensive and knowledge-intensive industries and pointed out that the application of AI technologies has led to a reduction in working hours and an improvement in efficiency within labor-intensive sectors. Moreover, the adoption of automated production lines and AI technologies has enabled the widespread integration of flexible working hours. Employees are increasingly able to personalize their work schedules based on individual preferences (Autor, 2015). This shift has not only boosted productivity but also enabled companies to provide more flexible job opportunities and facilitate work-life balance for employees (Choudhury et al., 2021). However, the impact of AI on working hours is not always positive. In some cases, the introduction of intelligent monitoring systems could increase work pressure and prolong working hours (Spencer, 2023; Wei et al., 2024). Furthermore, AI may intensify disparities in workload distribution. Low-skilled workers benefit from reduced workloads due to AI assistance, while high-skilled workers are required to perform more complex tasks, which may lead to longer working hours.

4.3.2. Workplace and forms of work

AI applications have also transformed the workplace and forms of work. Remote collaboration tools and intelligent management systems are eliminating the physical boundaries of the traditional workplace (Gkinko & Elbanna, 2023).AI platforms facilitate global collaboration by enabling real-time data analysis and supporting remote teamwork. Marikyan et al. (2022) found that AI-based digital assistants improve tele-workers' productivity, specifically, they observed positive correlations between workers' performance expectations, social communication, and trust with their satisfaction of these assistants. Furthermore, AI technologies also change workplace management styles. Einola and Khoreva (2023) point out that AI monitoring systems enhance employee efficiency and safety, but also introduce concerns regarding potential privacy infringements. It is necessary to balance efficiency and employee satisfaction when implementing new technologies. Meanwhile, AI technologies have also been introduced by companies to enhance work conditions. For example, robot mediators are used in recruitment to improve efficiency and reduce bias (Nørskov et al., 2022). AI adoption also fosters a more positive workplace atmosphere, as workers increasingly trust AI systems and interact with robots (Arias et al., 2023). Moreover, AI applications have also contributed to more positive working atmospheres through increased employee trust in AI systems and interactions with robots (Zhou et al., 2024). However, concerns about substitution by AI may also lead to workplace deterioration, such as workers unethical decision-making to secure job opportunities (Bai et al., 2025).

AI has facilitated the diversification of employment forms, such as contract workers, part-time workers, freelance workers and teleworkers. Firstly, it is easier for freelance and contract workers to engage into enterprise projects with the application of AI technologies in task assignment and performance evaluation. To optimize production costs between machines and labor, companies often adopt flexible hiring and firing strategies to adjust workforce quickly, thereby increasing short-term job opportunities (Fornino & Manera, 2022). Additionally, AI algorithms allow firms to predict labor demand and optimize labor allocation, further enhancing workforce flexibility (Brynjolfsson & McAfee, 2014). Secondly, the development of internet communication technologies and cloud

computing has facilitated remote work and created favorable conditions for flexible jobs (Choudhury et al., 2021). On the one hand, freelance work, such as content creation, translation, and data annotation, has become more efficient with the help of AI algorithms. These jobs are typically temporary and are matched on demand through platforms. On the other hand, the online platforms established by AI algorithms have fostered various forms of gig economy (Dabić et al., 2023), such as ride-hailing, food delivery, home services, and automated content creation. For example, ride-hailing and food delivery platforms employ intelligent dispatch systems and automated route planning to improve service efficiency (Zervas et al., 2017).

4.4. AI and wage inequality

Wages have been a major focus of labor market research, and the widespread adoption of AI has exacerbated wage inequality. We then systematically synthesize the impact of AI on wage inequality across skill levels, industries, and regions.

4.4.1. AI and wage inequality across skill levels

AI has changed the demand for workers with different skills, exacerbating income inequality. Low-skilled workers are more likely to be replaced by machines, resulting in stagnant real wages for the majority. In addition, unemployed workers face difficulties in re-employment, potentially causing a significant decline in wages (Lankisch et al., 2019). Using data from the 2015 Chinese Census, Jia et al. (2024) found that an increase in robot density exacerbates household wealth inequality, particularly affecting younger and less-educated workers, due to unequal employment opportunities and the immobility of human capital. Moreover, AI-driven industrial upgrading increases the demand for skilled labor, disadvantaging low-skilled workers and worsening wage inequality (Zhou & Tyers, 2019). In contrast, the premium for high-skilled labor is more pronounced, especially in skill-intensive industries, and the integration of AI further amplifies these premiums (Wang et al., 2024). Domini et al. (2022) find that automation and AI-related investments contribute to wage inequality within French firms, driven by disparities among workers within the same company rather than by differences across sectors, companies, and occupations. Notably, new employees with superior skills tend to receive higher salaries. Furthermore, AI exacerbates wage inequality by optimizing production processes and reducing management complexity. Technical and managerial positions benefit most from AI-driven wage growth, while wages in clerical and administrative positions remain relatively stagnant.

4.4.2. AI and wage inequality across industries

AI technology drives the mobility of production factors among industries, with the speed of adaptation to new tasks and machine substitution varying by industry, thus contributing to wage inequality across industries. The gap in bargaining power between labor-intensive industries and skill-intensive industries widens, further exacerbating wage inequality (Cheng et al., 2024). In labor-intensive industries such as manufacturing, agriculture, and some service industries, the application of AI typically thaks the form of automation and robots replacing traditional manual labor, which depresses wages. Jiang et al. (2024) found that the increased automation of capital inputs has widened the wage gap between labor assisted by automated machinery and labor supported by general capital. This effect is particularly pronounced in non-state-owned manufacturing firms and technology-intensive firms, where automation investment substantially influences the wage gap. In the service industry, AI reduces the bargaining power of low-education frontline employees, worsens income inequality, and shrinks the share of labor income within enterprises (Chen et al., 2024). However, some studies have shown that AI can reduce wage gaps within technology-intensive firms, especially in knowledge-intensive firms such as finance and information technology. AI technologies have the potential to enhance enterprise productivity and facilitate income growth for all employees (Yuan et al., 2024).

4.4.3. AI and wage inequality across regions

While AI technologies enhance productivity, they also widen regional wage disparities. Fu et al. (2021) found that increased robot adoption correlates with a significant rise in income inequality. Economically developed regions are better positioned to leverage AI, with productivity gains driving higher demand for skilled labor and resulting in wage increases. In contrast, less-developed regions, which rely on labor-intensive industries, experience the displacement effects of AI, leading to higher unemployment and wage declines. On the one hand, significant disparities exist in the acquisition of human capital between developed and less-developed regions. Developed regions have greater access to human capital, including better infrastructure and educational resources, enabling workers to adapt more rapidly to AI technologies and secure higher wages. Conversely, less-developed regions lack quality education and skills training, leading to wage stagnation and increased unemployment risks. Furthermore, developed regions tend to receive more policy support and capital investment, fostering AI research, enhancing productivity, and raising wages. Less-developed regions, however, suffer from insufficient investment and fail to benefit from technological progress, thereby exacerbating wage inequality.

4.5. AI and policy

AI has significantly impacted the labor market, posing new challenges for policymakers in addressing unemployment risks and improving workers' employability. Therefore, we will explore how to provide effective unemployment protection and promote re-employment training programs in the context of AI.

4.5.1. Unemployment security policy

The large-scale application of AI technologies in production has raised concerns about unemployment. As intelligent technologies

replace repetitive tasks, low-skilled and medium-skilled workers face a higher risk of job displacement, prompting the need for policymakers to reassess existing unemployment protection systems (Darıcı et al., 2024). In technology-intensive industries, traditional unemployment benefits are insufficient to meet the demands of labor markets. Moreover, AI amplifies workforce volatility, as companies often implement temporary layoffs to reduce operational costs, leading to frequent short-term unemployment (Benito & Hernando, 2008). The widespread use of robots has also intensified workers' job insecurity and diminished their job satisfaction, resulting in poorer mental health, particularly among those in low-skilled, routine positions (Abeliansky et al., 2024). This increases employment instability, underscoring the need for more flexible and sustainable unemployment protection policies.

4.5.2. Training and re-employment policy

In response to the job displacement and changes of skill demands due to AI technologies, training and reemployment policies are critical to mitigate unemployment. It is undeniable that the widespread adoption of AI raises skill requirements, and traditional vocational training is insufficient. Many companies have launched training programs to help workers adapt to new jobs (Ključnikov et al., 2023). Additionally, local governments have also established public training institutions to offer free vocational transition courses for the unemployed, helping to alleviate the adverse impacts of automation. However, the effectiveness of these programs diminishes as the pace of technological disruption accelerates (Schmidpeter & Winter-Ebmer, 2021). Successful re-employment policies must not only address current skill demands but also ensure long-term employment stability. Research shows that increased investment in education, training and active labor market programs enhances workers' resilience and mitigates the adverse effects of technological change (Grigoli et al., 2020).

5. Discussion

5.1. Limitation

Bibliometric research methods offer several advantages; however, they also have notable limitations.

First, the literature analyzed in this study was sourced from the Web of Science. Although widely used for its extensive coverage and high-quality resources, Web of Science is not likely to include all the important studies. Future research should integrate multiple databases to enhance the representativeness of the analysis.

Second, the literature search was based on predefined search terms related to AI and the labor market. Given the broad and multifaceted nature of these concepts, the selected keywords may not fully encompass all relevant studies, potentially overlooking keywords and introducing sample bias.

Third, this study focused primarily on English-language literature, which may neglect valuable contributions from non-English studies. Similarly, Chinese research on AI technologies offers important perspectives on their impact on the labor market, and its exclusion may lead to an incomplete understanding of AI's effects.

5.2. Further research

The impact of AI technologies on the labor market has become a central focus for both academia and policymakers. Previous studies have studied the impact of AI on employment, wages and occupational skills. The following are several critical questions that deserve future exploration.

First, most current research focuses on the quantity of jobs, with limited attention to job quality. The widespread adoption of AI has led to job transitions, yet its effects on job stability and security have been largely overlooked. Future research could explore these aspects in greater depth.

Second, the existing literature lacks an in-depth analysis of specific AI subfields. For example, generative AI and predictive AI exhibit substantial differences in their application scenarios, skill requirements and effects on the labor market. Generative AI (e.g., content creation and design) is expected to significantly disrupt employment in creative industries, while predictive AI (e.g., data analysis and decision optimization) may reshape tasks in traditional sectors. Future research could investigate how these technologies affect industry structures, skill demands, and propose strategies for career development.

Third, new forms of work are emerging with the continuous evolution of AI technologies, including flexible work and gig economies. The rapid growth of the platform economy has broadened workers' access to a wider range of job opportunities. While some studies have examined the rise of these new work forms (Chen et al., 2022; Graham et al., 2017), the mechanisms through which AI drives these changes remain underexplored. Future research could explore AI's role in flexible employment, its effects on income stability and job security for gig workers, and its implications for social welfare systems.

6. Conclusion

We conducted a bibliometric analysis of literature on AI and the labor market by searching the Web of Science, retrieving 1647 English-language articles published between 2007 and 2024. Using CiteSpace, we visualized publication trends, keyword co-occurrence, keyword clustering and burst analysis. Based on these analyses, we systematically synthesized the literature and presented representative perspectives on AI and the labor market. The main findings are as follows: (1) Since 2015, the number of publications on AI and the labor market has increased rapidly, with research topics expanding to include employment, wage inequality and the practical applications of AI technologies. (2) Bibliometric analysis emphasizes a significant focus on the effects of AI on

employment and skill demand. Keywords such as "digital workplace" "job insecurity" "job quality" and AI applications have emerged as critical research areas. (3) This study provides a detailed discussion of research areas, including the impact of AI on employment, skill, job quality, wage and policy. We also provide a detailed presentation of representative views. (4) Future research directions include examining how AI affects job quality and job stability, and exploring the differential impact of various AI technologies on the labor market.

CRediT authorship contribution statement

Zhou Huilin: Writing – review & editing, Writing – original draft, Visualization. **Wang Linhui:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization. **Cao Yutong:** Writing – review & editing, Supervision, Formal analysis, Data curation. **Li Jincheng:** Resources, Methodology, Data curation.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Declaration of Competing Interest

The authors have no relevant financial or non-financial interests to disclose.

The authors have no competing interests to declare that are relevant to the content of this article.

All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

The authors have no financial or proprietary interests in any material discussed in this article.

Data availability

Data will be made available on request.

References

Abeliansky, A. L., Beulmann, M., & Prettner, K. (2024). Are they coming for us? Industrial robots and the mental health of workers. *Research Policy, 53*(3), Article 104956.

Abrardi, L., Cambini, C., & Rondi, L. (2022). Artificial intelligence, firms and consumer behavior: A survey. *Journal of Economic Surveys, 36*(4), 969–991.

Acemoglu, D., & Autor, D. H. (2011). Skills, tasks and technologies: Implications for employment and earnings. *Handbook of Labor Economics, 4*, 1043–1171.

Acemoglu, D., & Restrepo, P. (2018). The race between man and machine: Implications of technology for growth, factor shares, and employment. *American Economic Review, 108*(6), 1488–1542.

Acemoglu, D., & Restrepo, P. (2019). Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives, 33*(2), 3–30.

Acemoglu, D., & Restrepo, P. (2020). Robots and jobs: Evidence from U.S. labor markets. *Journal of Political Economy, 128*(6), 2188–2244.

Acemoglu, D., & Restrepo, P. (2022). Tasks, automation, and the rise in U.S. wage inequality. *Econometrica, 90*(5), 1973–2016.

Acemoglu, D., Autor, D., Hazell, J., & Restrepo, P. (2022). Artificial intelligence and jobs: Evidence from online vacancies. *Journal of Labor Economics, 40*(S1), S293–S340.

Arendt, L., Gałecka-Burdziak, E., Núñez, F., Pater, R., & Usabiaga, C. (2023). Skills requirements across task-content groups in Poland: What online job offers tell us. *Technological Forecasting and Social Change, 187*, Article 122245.

Arias, M., Pérez, Rivero, C. A., & Carrero Márquez, O. (2023). Artificial intelligence to manage workplace bullying. *Journal of Business Research, 160*, Article 113813.

Audrin, B., Audrin, C., & Salamin, X. (2024). Digital skills at work – Conceptual development and empirical validation of a measurement scale. *Technological Forecasting and Social Change, 202*, Article 123279.

Autor, D. H. (2015). Why are there still so many jobs? The history and future of workplace automation. *Journal of Economic Perspectives, 29*(3), 3–30.

Bai, M., Zhang, H., Zhang, J., Jiang, Y., & Xu, J. (2025). Challenging or threatening? The double-edged sword effect of intelligent technology awareness on accountants' unethical decision-making. *Journal of Business Ethics, 197*, 159–175.

Benito, A., & Hernando, I. (2008). Labour demand, flexible contracts and financial factors: Firm-level evidence from Spain. *Oxford Bulletin of Economics and Statistics, 70*(3), 283–301.

Bessen, J., Goos, M., Salomons, A., & Berge, W. (2023). What happens to workers at firms that automate? *The Review of Economics and Statistics*, 1–45.

Bhargava, A., Bester, M., & Bolton, L. (2021). Employees' perceptions of the implementation of robotics, artificial intelligence, and automation (RAIA) on job satisfaction, job security, and employability. *Journal of Technology in Behavioral Science, 6*, 106–113.

Brynjolfsson, E., & McAfee, A. (2014). *The second machine age: Work, progress, and prosperity in a time of brilliant technologies*. New York: W.W. Norton & Company.

Camiña, E., Díaz-Chao, Á., & Torrent-Sellens, J. (2020). Automation technologies: Long-term effects for Spanish industrial firms. *Technological Forecasting and Social Change, 151*, Article 119828.

Chen, C., Frey, C. B., & Presidente, G. (2022). Automation or globalization? The impacts of robots and Chinese imports on jobs in the United Kingdom. *Journal of Economic Behavior Organization, 204*, 528–542.

Chen, H., Chen, C., Li, Y., Qin, L., & Qin, M. (2022). How Internet usage contributes to livelihood resilience of migrant peasant workers? Evidence from China. *Journal of Rural Studies, 96*, 112–120.

Chen, K., Chen, X., Wang, Z., & Zvarych, R. (2024). Does artificial intelligence promote common prosperity within enterprises? —Evidence from Chinese-listed companies in the service industry. *Technological Forecasting and Social Change, 200*, Article 123180.

Cheng, C., Luo, J., Zhu, C., & Zhang, S. (2024). Artificial intelligence and the skill premium: A numerical analysis of theoretical models. *Technological Forecasting and Social Change, 200*, Article 123140.

Cheng, C., Wang, L., Xie, H., & Yan, L. (2023). Mapping digital innovation: A bibliometric analysis and systematic literature review. *Technological Forecasting and Social Change, 194*, Article 122706.

Choi, T., & Leigh, N. G. (2024). Artificial intelligence's creation and displacement of labor demand. *Technological Forecasting and Social Change, 209*, Article 123824.

Choudhury, P., Foroughi, C., & Larson, B. (2021). Work-from-anywhere: The productivity effects of geographic flexibility. *Strategic Management Journal, 42*(4), 655–683.

Cortes, G. M., Jaimovich, N., & Siu, H. E. (2017). Disappearing routine jobs: Who, how, and why? *Journal of Monetary Economics, 91*, 69–87.
Dabić, M., Maley, J. F., Švarc, J., & Poček, J. (2023). Future of digital work: Challenges for sustainable human resources management. *Journal of Innovation Knowledge, 8*(2), Article 100353.
Damioli, G., Van Roy, V., Vertesy, D., & Vivarelli, M. (2023). AI technologies and employment: Micro evidence from the supply side. *Applied Economics Letters, 30*(6), 816–821.
Darıcı, S., Riaz, M., Demir, G., Gencer, Z. T., & Pamucar, D. (2024). How will I break AI? Post-Luddism in the AI age: Fuzzy MCDM synergy. *Technological Forecasting and Social Change, 202*, Article 123327.
Deming, D. J. (2017). The growing importance of social skills in the labor market*. *The Quarterly Journal of Economics, 132*(4), 1593–1640.
Domini, G., Grazzi, M., Moschella, D., & Treibich, T. (2021). Threats and opportunities in the digital era: Automation spikes and employment dynamics. *Research Policy, 50*(7), Article 104137.
Domini, G., Grazzi, M., Moschella, D., & Treibich, T. (2022). For whom the bell tolls: The firm-level effects of automation on wage and gender inequality. *Research Policy, 51*(7), Article 104533.
Duan, M., Hou, Y., Zhang, B., Chen, C., Sun, Y., Luo, Y., & Tan, T. (2024). Skill sets and wage premium: A network analysis based on Chinese agriculture online job offers. *Technological Forecasting and Social Change, 201*, Article 123260.
Durongkadej, I., Hu, W., & Wang, H. E. (2024). How artificial intelligence incidents affect banks and financial services firms? A study of five firms. *Finance Research Letters, 70*, Article 106279.
Egana-delSol, P., Bustelo, M., Ripani, L., Soler, N., & Viollaz, M. (2022). Automation in Latin America: Are women at higher risk of losing their jobs? *Technological Forecasting and Social Change, 175*, Article 121333.
Einola, K., & Khoreva, V. (2023). Best friend or broken tool? Exploring the co-existence of humans and artificial intelligence in the workplace ecosystem. *Human Resource Management, 62*(1), 117–135.
Feigenbaum, J., & Gross, D. P. (2024). Answering the call of automation: How the labor market adjusted to mechanizing telephone operation. *The Quarterly Journal of Economics, 139*(3), 1879–1939.
Fierro, L. E., Caiani, A., & Russo, A. (2022). Automation, job polarisation, and structural change. *Journal of Economic Behavior Organization, 200*, 499–535.
Fornino, M., & Manera, A. (2022). Automation and the future of work: Assessing the role of labor flexibility. *Review of Economic Dynamics, 45*, 282–321.
Frey, C. B., & Osborne, M. A. (2017). The future of employment: How susceptible are jobs to computerisation? *Technological Forecasting and Social Change, 114*, 254–280.
Fu, X., Bao, Q., Xie, H., & Fu, X. (2021). Diffusion of industrial robotics and inclusive growth: Labour market evidence from cross country data. *Journal of Business Research, 122*, 670–684.
Garg, S., Ahmad, A., & Madsen, D. (2024). Academic writing in the age of AI: Comparing the reliability of ChatGPT and Bard with Scopus and Web of Science. *Journal of Innovation Knowledge, 9*(4), Article 100563.
Giordano, V., Spada, I., Chiarello, F., & Fantoni, G. (2024). The impact of ChatGPT on human skills: A quantitative study on twitter data. *Technological Forecasting and Social Change, 203*, Article 123389.
Gkinko, L., & Elbanna, A. (2023). Designing trust: The formation of employees' trust in conversational AI in the digital workplace. *Journal of Business Research, 158*, Article 113707.
Gong, J., & Png, I. P. L. (2024). Automation enables specialization: Field evidence. *Management Science, 70*(3), 1580–1595.
Goos, M., Manning, A., & Salomons, A. (2014). Explaining job polarization: Routine-biased technological change and offshoring. *American Economic Review, 104*(8), 2509–2526.
Graham, M., Hjorth, I., & Lehdonvirta, V. (2017). Digital labour and development: Impacts of global digital labour platforms and the gig economy on worker livelihoods. *Transfer: European Review of Labour and Research, 23*(2), 135–162.
Gravina, A. F., & Pappalardo, M. R. (2022). Are robots in rich countries a threat for employment in emerging economies? *Economics Letters, 221*, Article 110888.
Grigoli, F., Koczan, Z., & Topalova, P. (2020). Automation and labor force participation in advanced economies: Macro and micro evidence. *European Economic Review, 126*, Article 103443.
Huang, Y. (2024). Digital transformation of enterprises: Job creation or job destruction? *Technological Forecasting and Social Change, 208*, Article 123733.
Jestl, S. (2024). Industrial robots, and information and communication technology: The employment effects in EU labour markets. *Regional Studies, 58*(11), 1981–1998.
Jia, N., Yan, X., & Yu, S. (2024). Does increasing robot density exacerbate wealth inequality? *Journal of Asian Economics, 95*, Article 101830.
Jiang, H., Wang, X., & Liu, C. (2024). Automated machines and the labor wage gap. *Technological Forecasting and Social Change, 206*, Article 123505.
Kim, D., Yeon, J., Kim, J. H., Kim, M., Song, Y., & Lee, D. (2023). Different effects of working hour reduction on labor-intensive and knowledge-intensive industries in the era of artificial intelligence: A meta-frontier approach. *Applied Economics, 55*(21), 2493–2504.
Ključnikov, A., Popkova, E. G., & Sergi, B. S. (2023). Global labour markets and workplaces in the age of intelligent machines. *Journal of Innovation Knowledge, 8*(4), Article 100407.
Lankisch, C., Prettner, K., & Prskawetz, A. (2019). How can robots affect wage inequality? *Economic Modelling, 81*, 161–169.
Li, X., Hui, E. C., Lang, W., Zheng, S., & Qin, X. (2020). Transition from factor-driven to innovation-driven urbanization in China: A study of manufacturing industry automation in Dongguan City. *China Economic Review, 59*, Article 101382.
Lin, J. (2011). Technological adaptation, cities, and New Work. *Review of Economics and Statistics, 93*(2), 554–574.
Lu, S. F., Rui, H., & Seidmann, A. (2018). Does technology substitute for nurses? Staffing decisions in nursing homes. *Management Science, 64*(4), 1842–1859.
Ma, H., Gao, Q., Li, X., & Zhang, Y. (2022). AI development and employment skill structure: A case study of China. *Economic Analysis and Policy, 73*, 242–254.
Mann, K., & Püttmann, L. (2023). Benign effects of automation: New evidence from patent texts. *Review of Economics and Statistics, 105*(3), 562–579.
Marikyan, D., Papagiannidis, S., Rana, O. F., Ranjan, R., & Morgan, G. (2022). "Alexa, let's talk about my productivity": The impact of digital assistants on work productivity. *Journal of Business Research, 142*, 572–584.
Nguyen, Q. P., & Vo, D. H. (2022). Artificial intelligence and unemployment: An international evidence. *Structural Change and Economic Dynamics, 63*, 40–55.
Nørskov, S., Damholdt, M. F., Ulhøi, J. P., Jensen, M. B., Mathiasen, M. K., Ess, C. M., & Seibt, J. (2022). Employers' and applicants' fairness perceptions in job interviews: Using a teleoperated robot as a fair proxy. *Technological Forecasting and Social Change, 179*, Article 121641.
Pan, M., Li, D., Wu, H., & Lei, P. (2024). Technological revolution and regulatory innovation: How governmental artificial intelligence adoption matters for financial regulation intensity. *International Review of Financial Analysis, 96*, Article 103535.
Qin, X., Xu, W., Chen, H., 'Caron,' Zhong, J., Sun, Y., & Li, X. (2022). Automation, firm employment and skill upgrading: Firm-level evidence from China. *Industry and Innovation, 29*(9), 1075–1107.
Robinson, S., Orsingher, C., Alkire, L., De Keyser, A., Giebelhausen, M., Papamichail, K. N., & Temerak, M. S. (2020). Frontline encounters of the AI kind: An evolved service encounter framework. *Journal of Business Research, 116*, 366–376.
Sachan, S., Almaghrabi, F., Yang, J.-B., & Xu, D.-L. (2024). Human-AI collaboration to mitigate decision noise in financial underwriting: A study on FinTech innovation in a lending firm. *International Review of Financial Analysis, 93*, Article 103149.
Schmidpeter, B., & Winter-Ebmer, R. (2021). Automation, unemployment, and the role of labor market training. *European Economic Review, 137*, Article 103808.
Sen, W., Hong, Z., & Xiaomei, Z. (2022). Effects of human–machine interaction on employee's learning: A contingent perspective. *Frontiers in Psychology, 13*, Article 876933.
Spencer, D. A. (2023). Automation and well-being: Bridging the gap between economics and business ethics. *Journal of Business Ethics, 187*(2), 271–281.
Stemmler, H. (2023). Automated deindustrialization: How global robotization affects emerging economies—Evidence from Brazil. *World Development, 171*, Article 106349.
Tyson, L. D., & Zysman, J. (2022). Automation, AI & work. *Daedalus, 151*(2), 256–271.
Wang, C., Zheng, M., Bai, X., Li, Y., & Shen, W. (2023). Future of jobs in China under the impact of artificial intelligence. *Finance Research Letters, 55*, Article 103798.

Wang, L., Zhou, H., Qian, Y., & Dong, Z. (2024). Robot application shocks, occupational transferable skills and occupational-skill desirability. *Management World, 11*, 85–104.
Wang, S., Wang, Y., & Li, C. (2024). AI-driven capital-skill complementarity: Implications for skill premiums and labor mobility. *Finance Research Letters, 68*, Article 106044.
Wei, X., Xu, J., & Cao, H. (2024). Production automation upgrades and the mystery of workers' overwork: Evidence from a manufacturing employer-employee matching survey in China. *Journal of Asian Economics, 91*, Article 101711.
Xie, M., Ding, L., Xia, Y., Guo, J., Pan, J., & Wang, H. (2021). Does artificial intelligence affect the pattern of skill demand? Evidence from Chinese manufacturing firms. *Economic Modelling, 96*, 295–309.
Yuan, Y., Sun, Y., & Chen, H. (2024). Does artificial intelligence affect firms' inner wage gap? *Applied Economics*, 1–7.
Zervas, G., Proserpio, D., & Byers, J. W. (2017). The rise of the sharing economy: Estimating the impact of Airbnb on the hotel industry. *Journal of Marketing Research, 54*(5), 687–705.
Zhang, X., Sun, M., Liu, J., & Xu, A. (2024). The nexus between industrial robots and employment in China: The effects of technology substitution and technology creation. *Technological Forecasting and Social Change, 202*, Article 123341.
Zhang, Y., Zhang, M., Li, J., Liu, G., Yang, M., & Li, S. (2021). A bibliometric review of a decade of research: Big data in business research – Setting a research agenda. *Journal of Business Research, 131*, 374–390.
Zhou, Q., Chen, K., & Cheng, S. (2024). Bringing employee learning to AI stress research: A moderated mediation model. *Technological Forecasting and Social Change, 209*, Article 123773.
Zhou, Y., & Tyers, R. (2019). Automation and inequality in China. *China Economic Review, 58*, Article 101202.